

Database Management Systems (DBMS) – Core Concepts

Schema vs. Instance

- **Schema:** The overall design of the database (like a blueprint).
- **Instance:** The actual data stored at a particular moment.

ACID Properties

- **Atomicity:** Transactions are all-or-nothing.
- **Consistency:** Database moves from one valid state to another.
- **Isolation:** Concurrent transactions don't interfere.
- **Durability:** Once committed, data persists even after failures.

Concurrency Control

- Ensures **isolation** by locking or timestamp ordering.
- Maintains **consistency** by preventing conflicts like lost updates or dirty reads.

Artificial Intelligence – Core Concepts

Machine Learning Paradigms

- **Supervised Learning:** Learn from labeled data (e.g., classification).
- **Unsupervised Learning:** Discover hidden patterns (e.g., clustering).
- **Reinforcement Learning:** Learn by trial and error with rewards.

Neural Networks

- **Input Layer:** Features.
- **Hidden Layers:** Transformations via activation functions.
- **Output Layer:** Predictions or classifications.

Key AI Applications

- **Natural Language Processing (NLP):** Chatbots, translation.
- **Computer Vision:** Image recognition, OCR.
- **Expert Systems:** Rule-based decision making.